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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2501-1 – CONSUMER ELECTRONICS
(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Sound is a _____ Wave
A) Transverse
B) Travelling
C) Longitudinal
D) Progressive
2. The velocity of sound is largest in _____
A) air
B) steel
C) Vacuum
D) water
3. What do we call the distance between two consecutive compressions of a sound wave?
A) Wave number
B) Wavelength
C) Frequency
D) Amplitude
4. When sound strikes on only one surface of the diaphragm the mike is said to be _____
A) pressure gradient microphone
B) pressure type microphone
C) velocity microphone
D) electret microphone
5. Acoustic doublet action can be prevented by using
A) a baffle
B) acoustic coupling
C) an enclosure
D) acoustic decoupling
6. In a speaker when voice coil moves in the magnetic gap it acts as _____
A) Motor
B) Generator
C) Amplifier
D) Compensator
7. The other name for baffle is _____
A) Cabinet
B) network
C) crystal
D) mixer
8. In disc recording needle is known as _____
A) Track
B) Armature
C) Stylus
D) Magnet
9. _____ is used to convert the vibration of playback stylus into electrical signal
A) Amplifier
B) Motor
C) Cartridge
D) Stylus
10. Cartridge functions as a _____
A) Loudspeaker
B) Stylus
C) Amplifier
D) Microphone
11. The wiper rotates in one plane only, so this type of electro-mechanical switch is called
A) Two motion selector
B) Uniselecto
C) Single motion selector
D) Biselecto

12. The step by step exchange is called as
 A) Relay exchange
 B) Strowger exchange
 C) Final exchange
 D) multi exchange
13. PABX switching stands for
 A) Private automatic branch exchange.
 B) Primary automatic branch exchange.
 C) Public automatic branch exchange.
 D) Principle automatic branch exchange.
14. A circuit switching network
 A) Blocks excess traffic.
 B) Provides longer delivery times
 C) Normally accepts all traffic
 D) Provides shorter delivery times
15. Heat always travels from _____
 A) Cooler area to warmer area
 B) Warmer area
 C) Warmer area to cooler area
 D) Cooler area
16. _____ obeys the reciprocity law
 A) selenium
 B) carbon
 C) silicon
 D) graphite
17. Refrigerants are _____ mediums
 A) Filtration
 B) Pressure
 C) Freezer
 D) Heat carrying
18. Fax can be transmitted over normal _____
 A) Wirelessly
 B) Telephone lines
 C) Air
 D) High tension cable
19. Manipulator or Rover is _____
 A) Main body of robot
 B) Brain of robot
 C) Muscles of robot
 D) cerebellum of robot
20. In a washing machine _____ at the bottom circulates water upwards in large circles
 A) Power cord
 B) Front control panel
 C) Pulsator disk
 D) Lint filte

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the construction of LCD with relevant diagram.	8	L*2	2	1
	b) Describe the characteristics of microphones and audio amplifier briefly.	8	L2	1	1
2.	a) Explain the construction and working of plasma display	8	L2	2	1
	b) Explain the construction and working of direct radiating dynamic cone speaker	8	L2	1	1
3.	a) Explain the traffic control workflow using CCTV.	8	L2	2	1
	b) Explain the following terms i. Doppler effect ii. Sound Intensity iii. Superposition of Waves iv. Beats	8	L2	1	1
Unit – II					
4.	a) Explain the two methods of sound recording with help of neat diagram.	8	L2	3	1
	b) Explain the working of CD player with block diagram.	8	L2	4	1
5.	a) With the help of a neat block diagram, explain the Disc recording.	8	L2	3	1
	b) Explain different modulations, it's need and advantages.	8	L2	4	1

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|-------|---|---|----|---|---|
| 6. a) | Explain different topologies of Local Area Networks (LANs) with the help of neat diagrams and list the advantages of LAN. | 8 | L2 | 4 | 1 |
| b) | List the difference between Disc and Magnetic Recording. | 4 | L2 | 3 | 1 |
| c) | List and explain any two distortion of sound in HiFi systems. | 4 | L2 | 3 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Explain in detail the refrigeration cycle. | 8 | L2 | 5 | 1 |
| b) | Explain the basic fax machine operation with the block diagram. | 8 | L2 | 5 | 1 |
| 8. a) | How electronic ignition system works. Explain with a neat circuit diagram. | 8 | L2 | 5 | 1 |
| b) | Explain the working principle of microwave oven with neat diagram. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
